

Askademia - Educational Q&A Forum

Year
Accomplished
2025

Role
Personal Project -
Fullstack Developer

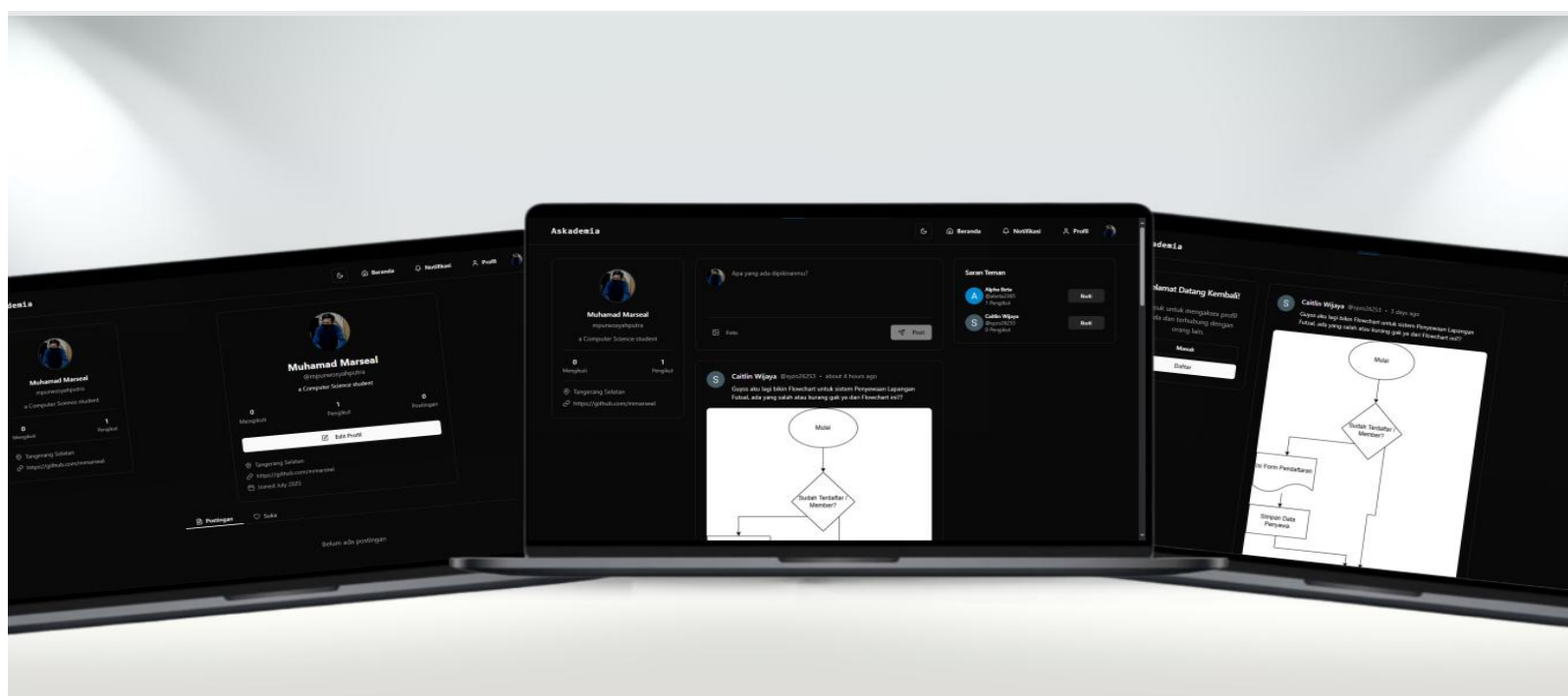
Publication Link
<https://askademia-chi.vercel.app/>

Project Description

As a university student, I often struggled to find answers for my assignments. I wanted to be able to discuss topics with many students, not just my campus friends. However, my search often ended in "noisy" general forums where quality academic discussions were frequently drowned out by irrelevant content. For solving coding errors, I was helped by Stack Overflow. On the other hand, I was also an active Quora user for finding interesting insights from people's experiences and discussions. I was greatly helped by the efficiency of Stack Overflow and the quality of discussions on Quora. From that experience, I began to realize something and a simple question arose: "Why isn't there one place that can truly be a study buddy? What if there was a space that could be a learning companion for anyone, but was focused on helping students in an academic context?" From that frustration and inspiration, the idea to build Askademia was born.

At the time, I was diving deep into Next.js and wanted to immediately apply my knowledge to a real project. Askademia became my canvas. I challenged myself to build it full-stack, integrating Prisma and PostgreSQL for the database, and Tailwind CSS for styling. As an additional experiment, I tried using Clerk for authentication and user management in this project. My challenge was how to integrate the architecture, from the posting system, feedback messages, to authentication, so they wouldn't overlap, and to design this learning environment to stay true to the Askademia project idea. My focus was on writing solid code and ensuring every component functioned well.

The result is a fully-functional web app, proving that a focused solution can be very effective. For me, Askademia was an invaluable end-to-end exercise. Through this project, I not only managed to master a modern full-stack workflow, from designing a database schema to deployment, but I also learned how to take an idea and execute it into a real, tangible digital product. This project sharpened my problem-solving skills in a very practical way.



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DapurIbu: A Fullstack Mobile Recipe App for Daily Inspiration

Year Accomplished
2025

Role
Personal Project - Fullstack Mobile

Publication Link

- Demo App:

<https://youtube.com/shorts/AX6YKciPX7E?si=eyl2kaP7LVfwEum>

- Expo Go:

<https://expo.dev/accounts/marseal/projects/mobile>

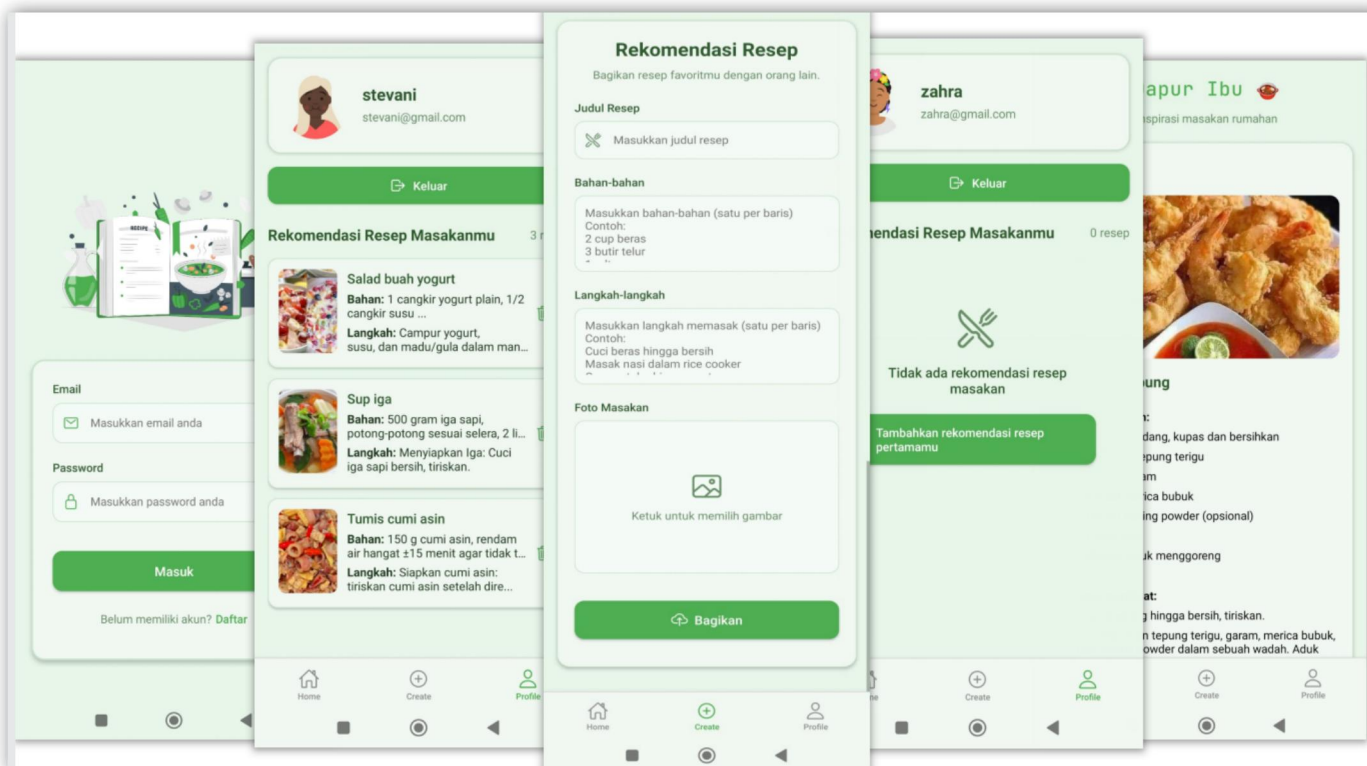
- APK Download (Android):

<https://drive.google.com/file/dapurlbu-mobile.apk>

Project Description

This project was born from a simple problem at home, where my mother would often get confused and tired thinking about the menu for the next day. Inspired by this real-world need, I wanted to create a practical digital solution that could help not only my mother but also many others facing a similar dilemma. And so, DapurIbu was born: a mobile application to easily share and discover recipe inspiration.

As my first mobile project after diving into web development with Next.js, I chose React Native to stay within the JavaScript ecosystem. I also made a conscious decision to use Expo, which allowed me to develop on a physical device without relying on a heavy Android Studio emulator. This was a smart solution to my hardware limitations, turning a hurdle into an opportunity to learn a more efficient development workflow. I faced the challenge of designing an efficient API, having to map out all the mobile app's data needs while keeping the code clean and functional in every part. Meanwhile, on the mobile app, I learned to bridge the communication between the front-end and back-end, which included managing state to ensure user data and recipe lists remained in sync across various screens for a smooth and intuitive user experience. The result is DapurIbu, a fully functional mobile application where users can create an account, log in, upload recipes complete with photos, browse recipes from others, and experience seamless interaction between the backend and the mobile client. I gained a deeper understanding of client-server architecture. For me, DapurIbu wasn't just a technical project, but a learning journey of how to start from a personal problem, formulate it into a solution, and then execute it, even with limitations.



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NotePal - Your Smart Lecture Companion

Year Accomplished
2025

Role
Project Leader (Team of 6) - Class Assignment

Publication Link
<https://drive.google.com/drive/folders/>

Project Description

NotePal is an AI and cloud-based smart lecture note-taking application capable of recording audio, transcribing it to text, saving notes to the cloud, and automatically providing smart summary and highlight features. This project was developed within the framework of a startup simulation that my friends and I created to a fulltask from our lecturer. I initiated this project after seeing a real problem around me, where many of my friends struggled to take notes during class. As a university student who also faces similar challenges, I felt driven to create a solution that would not only help myself but also benefit others.

We began by conducting simple research: observing friends' behaviors, asking about their difficulties, and then formulating a simple yet applicable solution idea. Through this, we discovered that the problem wasn't just the difficulty of taking notes during class, but also the issue of scattered and disorganized material, both written and digital as well as problems with storage and data loss. In this startup simulation project, I took on the role of Project Leader for a team of six. I led the process from the initial brainstorming, delegated research and design tasks, and planned business aspects such as estimating application development costs for servers, cloud, maintenance, security, and data backup. Furthermore, I prepared a development strategy in the form of a roadmap, starting from an MVP that only included basic note and reminder features, to a plan for advanced features like collaborative notes, file sharing, and even the AI summarizer. The biggest challenge for me was balancing the technical aspects with non-technical ones, such as budgeting, server cost simulation, and calculating team needs. This process taught me a lot about perspectives beyond coding namely, how an idea can evolve into a real product that is both feasible and sustainable. I facilitated intensive discussions to ensure our team had a shared understanding in designing a solid product flow.

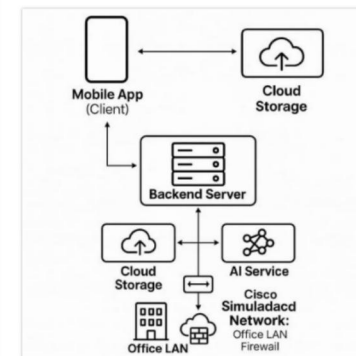
The final result is a mature product concept, which we poured into a comprehensive business proposal, a final presentation, and design mockups created using AI. Although it is not yet a functional application, this project provided me with two important lessons. First, as a leader, I learned the vital importance of clear communication to keep the team aligned on complex goals. Second, I gained direct experience with the entire product development lifecycle: from idea and feature design to business considerations. This was my first experience thinking not just as a student, but as a product planner.

Project 3 of 5

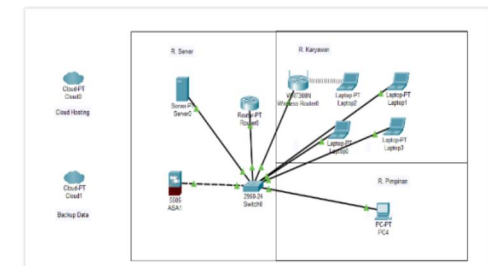
Tampilan Aplikasi (Mockup UI)



Arsitektur Aplikasi



Infrastruktur Jaringan



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30-Day Coding Challenge & Problem-Solving Certification

Year Accomplished
2025

Role
Personal Skill Development

Publication Link
<https://github.com/mmarseal/coding-challenges>

Project Description

I realized that behind every great application lies a strong foundation in problem-solving. To proactively build that foundation outside of my formal curriculum, I dedicated myself to a challenge focused on honing my skills in algorithms and data structures.

I began this journey with a 30-day challenge on HackerRank, practicing consistency by solving a problem each day and documenting the process in my GitHub repository. As shown in my documented thought process, I broke down each challenge into several steps: understanding the problem, creating pseudocode, and then implementing it into code. This journey also taught me about resilience. The challenge was briefly paused as I needed to prioritize my family and focus on caring for my mother while she was hospitalized. However, after things improved, I returned to the challenge with an even stronger commitment. To validate my understanding, I successfully obtained the "Problem Solving (Basic)" certificate, which I see not as a finish line, but as a starting point to continue tackling more complex challenges on platforms like HackerRank and LeetCode in the future.

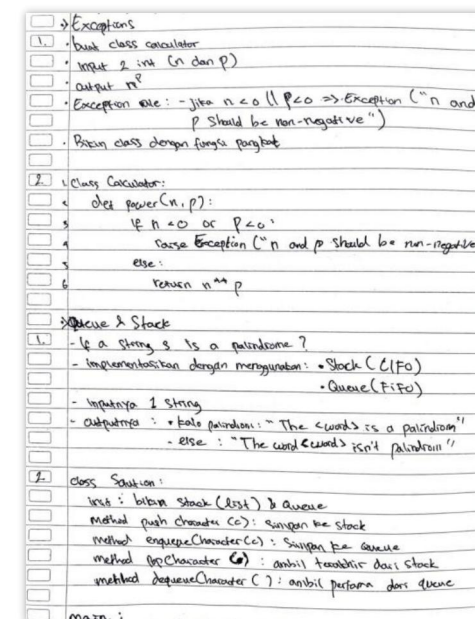
The outcome of this challenge isn't a product, but measurable skill growth. I became more skilled at breaking down complex problems into logical and efficient steps. Most importantly, my understanding and application of fundamental Data Structures and Algorithms (DSA) became much stronger. This is the foundation I believe is essential for building more reliable and high-performance applications.



Certificate Hackerrank

Name	Last commit message	Last commit date
..		
day00_helloWorld.py	day 0: I/O [HackerRank]	2 months ago
day01_dataTypes.py	Refactor: DayX -> DayXX	3 days ago
day02_operators.py	Refactor: DayX -> DayXX	3 days ago
day03_conditionalStatements.py	Refactor: DayX -> DayXX	3 days ago
day04_classVsInstance.py	Refactor: DayX -> DayXX	3 days ago
day05_loops.py	Refactor: DayX -> DayXX	3 days ago
day06_evenStr_OddStr.py	Refactor: DayX -> DayXX	3 days ago
day07_arrays.py	Refactor: DayX -> DayXX	3 days ago
day08_DictionariesMaps.py	Refactor: DayX -> DayXX	3 days ago
day09_recursion.py	Refactor: DayX -> DayXX	3 days ago
day10_binaryNumbers.py	Day 10: Binary Numbers	last month
day11_2DArrays.py	Day 11: 2D Arrays	last month
day12_hashes.py	Day 12: Hashes	last month

Repository Github



My Problem-Solving Process

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WartegKu: UI/UX Design Concept for Local Eateries

Year
Accomplished
2025

Role
Personal Project -
Product Designer

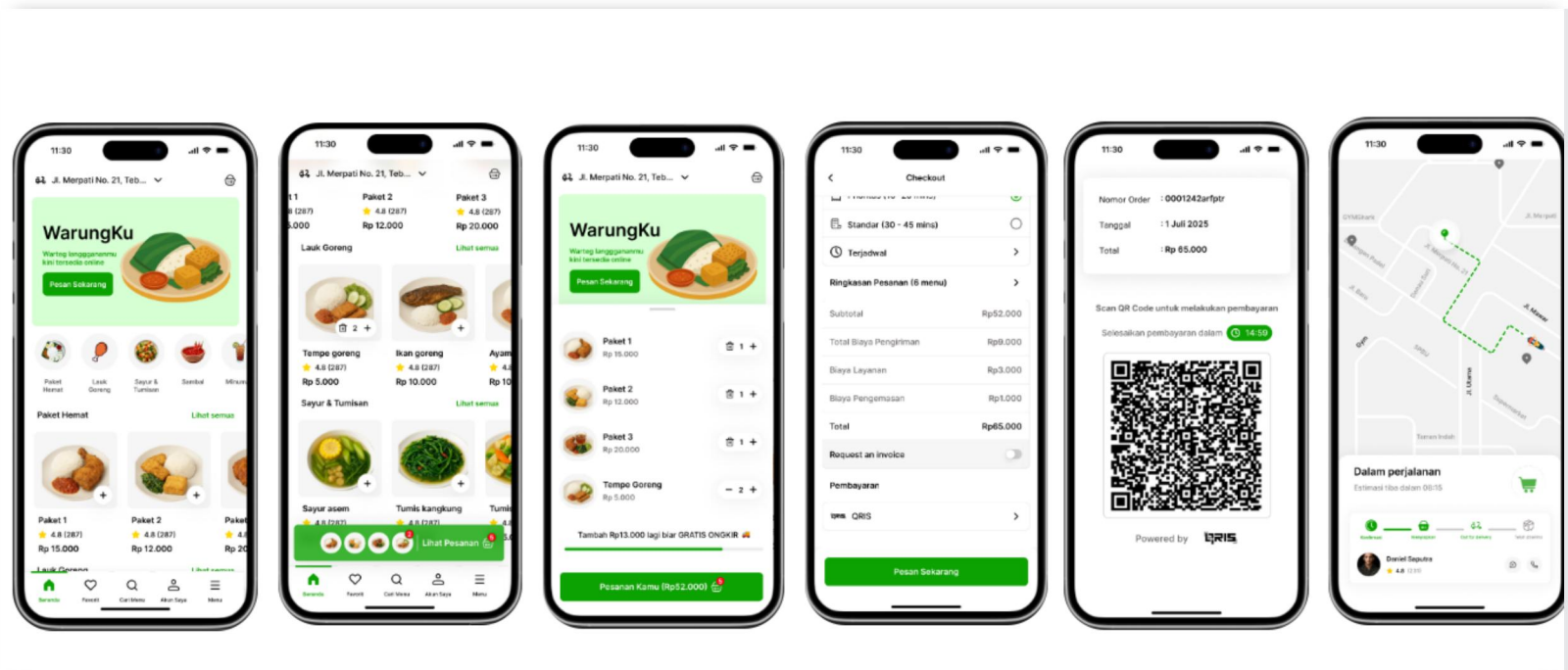
Publication Link
<https://www.figma.com/community/file/>

Project Description

As a university student, wartegs (local food stalls) are an essential part of my daily life. However, I often wished I could order food from my favorite stall without having to leave my busy campus schedule. From this personal experience, I saw an opportunity, *"what if the convenience of modern delivery apps could be applied to support local wartegs and MSMEs?"* To answer that question, I began designing 'WartegKu' in Figma. My process started by defining a Minimum Viable Product (MVP), the most essential features that would allow a user to view the menu, place an order, and complete the transaction. From there, I mapped it out into a simple and logical user flow.

This project became an opportunity for me to apply and deepen my understanding of UI/UX design principles. However, my biggest learning wasn't about the technical use of Figma, but a growth in my design mindset. I learned that even though I was focused on designing the customer-facing interface, I had to constantly consider the needs of the warteg owner. For example, I designed the menu display to be simple, imagining how easy the system would need to be for an owner to update their daily menu. This process of balancing the customer's desires with the operational ease for the owner truly changed how I view design; it's no longer just about visuals, but about being a bridge solution for the entire ecosystem.

Finishing the "WartegKu" design made me realize one thing, I truly enjoy the entire process, from identifying a problem and designing a solution to visualizing it. However, I don't want to stop at the design level. My aspiration is to have the technical ability to actually build an application like this into a real product that can help real warteg owners. This is the main reason why I am so eager to join the Apple Developer Academy: to gain the end-to-end development skills needed to turn empathy-driven ideas into functional solutions that benefit the community.



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